

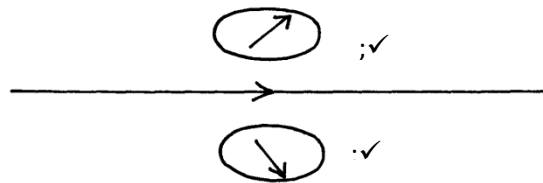
232/2

PHYSICS

PAPER 2

MARKING SCHEME

1.



2. The charged point conductor repels positive ions of the air while negative ions are attracted; ✓
 The positive ions dnft. Towards the flame forming an electric wind which drags the flame with it; ✓

3. Metre bridge has no zero error as it depends on balancing point; ✓

4. Frequency of the source; ✓

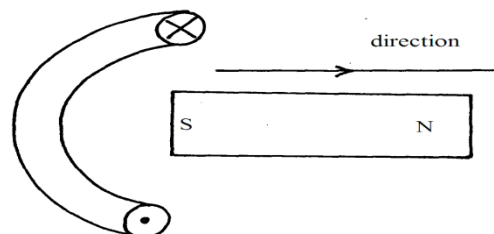
5. Total units. $= \frac{1200}{1000} \times 30 = 36 \text{Kwh};$ ✓

Cost $= 36 \times 8$

$= \text{Ksh } 288;$ ✓

6. Images are virtual; ✓ or upright or erect any 1 mark.

7.



8. a) Peak value $= 4 \text{cm} \times \frac{100}{1000}; =$ ✓

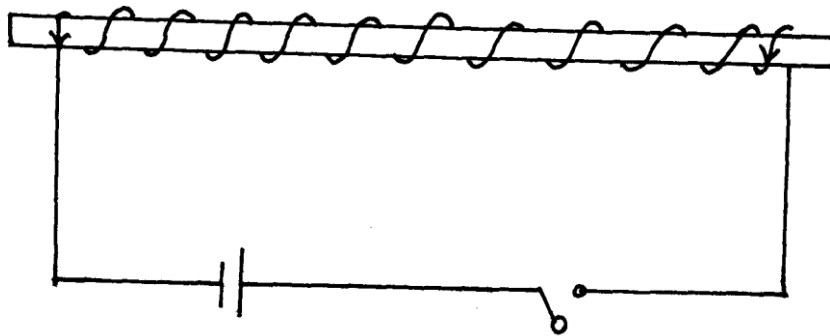
$= 0.4 \text{V};$ ✓

b) Period, $T = 8 \times \frac{0.8s}{1000} = 0.0064s$; ✓

frequency, $f = \frac{1}{0.0064} = 156.25Hz$; ✓

9. Intrinsic semi-conductor -their conductivity is enhanced by temperature only; ✓
 Extrinsic semi conductor -Their conductivity is enhanced by doping ; ✓
 An example of intrinsic is silicon and germanium; ✓ ½
 An example of extrinsic is P-type semi-conductor and n-type semi-conductor; ✓ ½

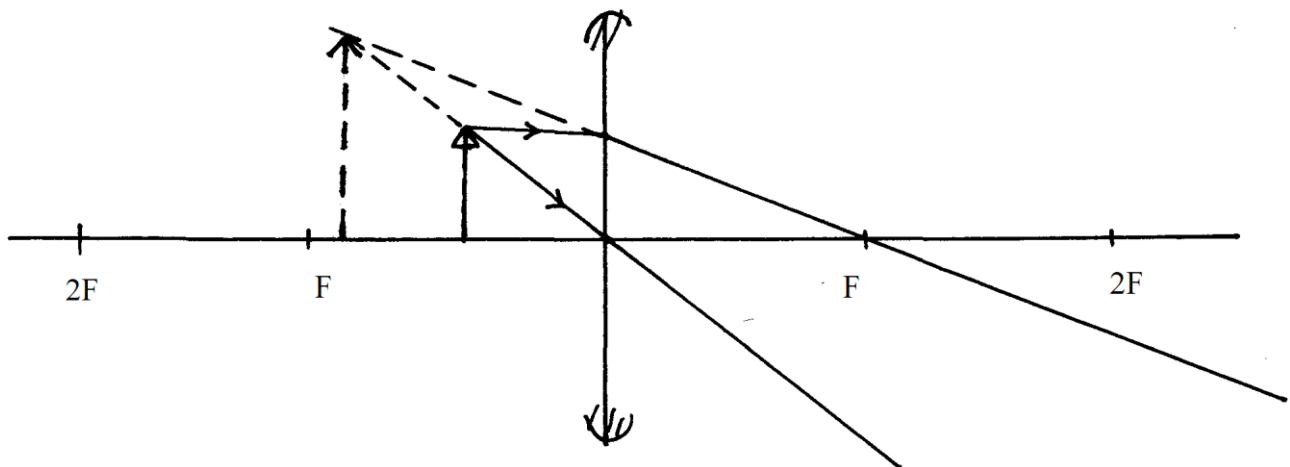
10.



; ✓ Correct circuit connection by closing the switch current flow as shown and AB becomes magnetised

11. Angle of incidence = Angle of reflection; ✓

12.



13. A is the mass number while z is the atomic number; ✓

14. a) Hard X-rays; ✓
 b) i) A focusing cathode, B-Electron beam, C-copper anode, D- lead shield ; ✓ ½ each.
 ii) Filament is heated it heats the cathode and electrons are emitted thermionically; ✓ emitted electrons are accelerated to the target by accelerating potential; ✓ upon hitting the target, X-rays are produced; ✓
 iii) The moving electrons possesses kinetic energy; ✓ most of this energy is converted into heat when the electrons meet the target.
 iv) X-rays are able to ionise the air particles; ✓ ionises air particles are used to discharge cloth materials to remove static charges; ✓ the static charges on the cloth can cause fire outbreak; ✓



v) $\lambda = \frac{C}{f}$

$$\frac{3.0 \times 10^8}{1.0 \times 10^{+16}} = 1.0 \times 10^{-11} \text{ m}; \checkmark$$

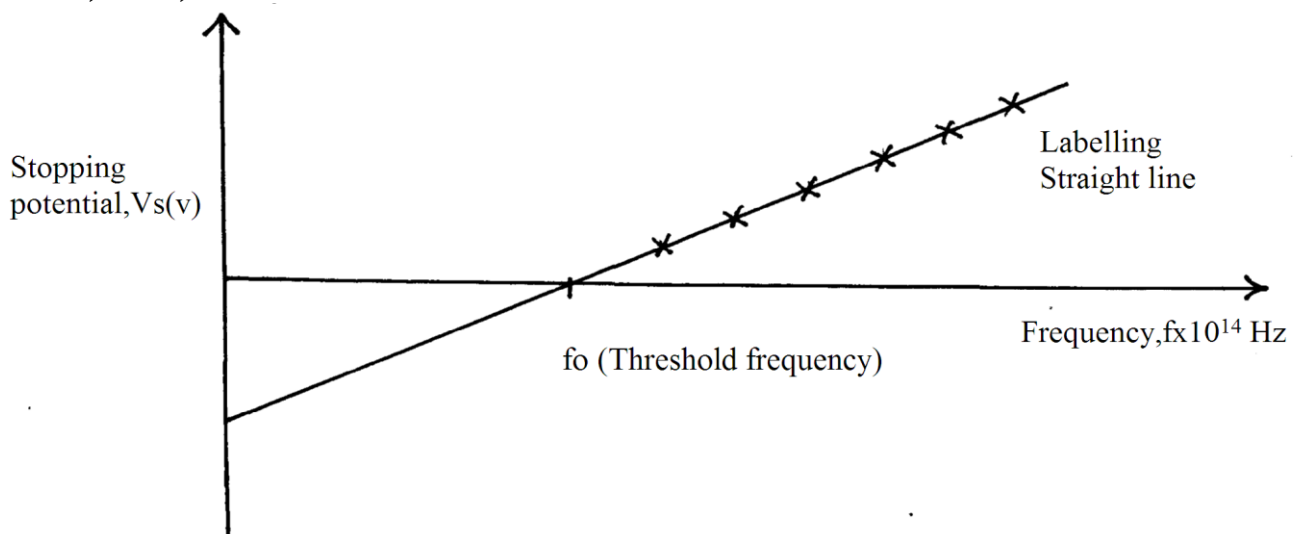
Also $\lambda = \frac{C}{f}$

$$\frac{3.0 \times 10^8}{1.0 \times 10^{+19}} = 1.0 \times 10^{-11} \text{ m}; \checkmark$$

Range $\lambda \pm 5.1 \times 10^{-11} \text{ m}$ to $1.0 \times 10^{-8} \text{ m}; \checkmark$

15. a) Is the process by which electrons are emitted from metal surface when irradiated with UV radiations of sufficient energy; $\checkmark$
- b) i) To remove impurities from the metal surface for photo electric emission; $\checkmark$
- ii) Gold leaf electroscope diverges / rises; $\checkmark$
 -Electrons are emitted from zinc surface which make electrons to move from gold leaf and stem making them positive and like charges repel; $\checkmark$ electrons become positively charged; $\checkmark$

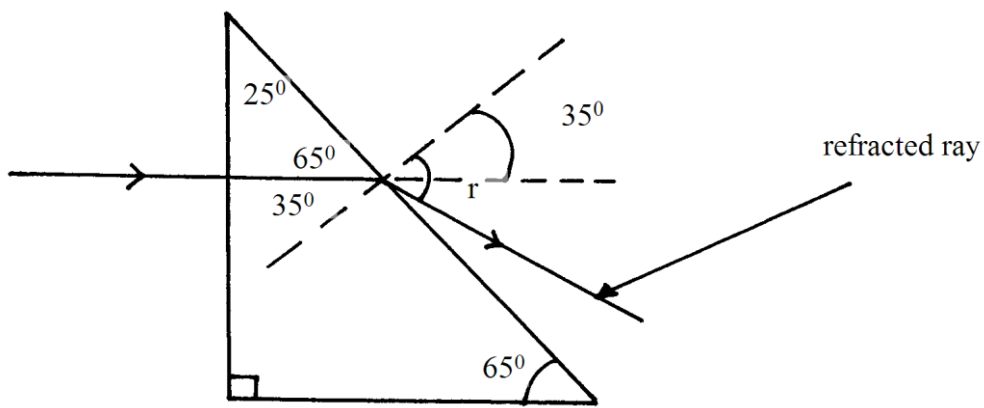
c) i)



ii) Stopping potential is the potential at the anode that completely stops the electrons from getting to it; $\checkmark$

d) Max K.E = eVs; $\checkmark$
 $= 1.6 \times 10^{-19} \times 0.5$; $\checkmark$
 $= 1.2 \times 10^{-19}$; $\checkmark$

16. a) Ratio of sine of angle of incidence in air to sine of angle of refraction in denser medium is always a constant; $\checkmark$
- b) i)

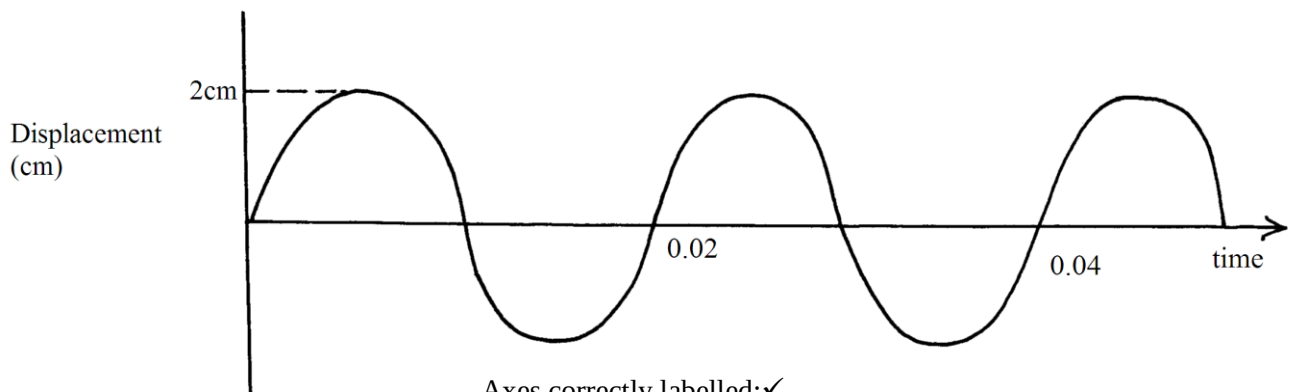


ii) $\frac{\sin i}{\sin r} = n$; ✓
 $\frac{\sin 25^\circ}{\sin r} = 1.5$; ✓
 $r = 59.4^\circ$; ✓

c) Angle of deviation $= 59.4 - 35$; ✓
 $= 24.4^\circ$; ✓

d) is because angle of incidence from denser glass is less than the critical angle for glass; ✓

17. a)



Axes correctly labelled; ✓
 2 complete circles; ✓
 $T = 1/50 = 0.02$; ✓ correct time indicated; ✓

b) Electromagnetic waves do not require a material medium ; ✓ while mechanical waves require material medium for their transmission; ✓

c) $V = \lambda f$; ✓ $2d = V_{xt}$; ✓
 $= 21 \times 10^3 \times 7.5 \times 10^{-2}$ $d = \frac{1575 \times 0.4}{2}$
 $= 1575 \text{ m/s}$; ✓ 315 m ; ✓

18. a) Iron keeps from closed loops; ✓ it makes the end of the bar magnets not to lose their magnetism by self demagnetisation; ✓



- b) when a magnet is being magnetised, the dipoles are arranged in the domains facing in all directions; ✓ as magnetising process continues the dipole in all the domain aligns themselves in same direction such that further magnetisation has no effect; ✓
- c)
 - i) When current is switched on electromagnet A becomes magnetised; ✓ B get attracted towards A and the horn makes sound; ✓ circuit is broken at C, Electromagnet loses magnetism and contact is again made, process repeats itself; ✓
 - ii) By increasing amount of current in circuit; ✓
By using a U-shaped electromagnet; ✓